

CSCE 753 COMPUTER VISION · MODULE 09 · LECTURE NOTES

Detection and Segmentation

Where, not just what — and how to measure it.

MODULE OUTCOMES

- Explain the detection problem and the standard architectures.
- Explain non-maximum suppression and its parameters.
- Compute and interpret mean average precision correctly.
- Distinguish semantic, instance, and panoptic segmentation.
- Choose a metric that matches the actual use.

1 Detection

DETECTION IS HARD BECAUSE THE OUTPUT HAS NO FIXED SIZE

Classification outputs one label per image. Detection outputs a variable number of boxes with labels — and a neural network produces fixed-size tensors. **That mismatch is the structural difficulty of the task**, and recognising it makes the architectural zoo comprehensible.

Every architecture is a way of handling the mismatch. Anchor-based methods predict offsets and a class from a fixed dense grid of candidate boxes; centre-based methods predict a box from each spatial location; query-based methods use a fixed set of learned query vectors. **In all cases the network's output is fixed-size** and the variability is handled afterwards.

All three produce far too many predictions and then prune them — which is what non-maximum suppression does (§2) — and the one exception is instructive.

DETR removed the pruning step by training with a bipartite matching loss: each prediction is matched to at most one ground-truth object by a Hungarian assignment (**CSCE 669 Module 11's matching problem, inside a loss function**), so duplicates are penalised during training rather than removed after it. **The architectural novelty was really a loss novelty**, which is worth noticing because it is a recurring pattern: the hard part of a structured-output problem is usually the loss, not the network.

Family	How it works	Trade
Two-stage	Propose candidate regions, then classify and refine each one.	Most accurate, slower. Faster R-CNN and its descendants.
One-stage anchored	Dense class and box-offset predictions over a fixed set of anchor boxes at every location.	Fast; the anchor shapes and scales are fiddly hyperparameters. RetinaNet, the YOLO family.

Family	How it works	Trade
Anchor-free	Predict a box directly from each centre location, with no anchor priors.	Simpler and with fewer hyperparameters , at comparable accuracy. FCOS, CenterNet.
Query-based	A fixed set of learned queries attends to image features; trained with a matching loss.	No non-maximum suppression at all , and notably slow to converge in training. DETR and successors.

Class imbalance is the specific difficulty of one-stage detectors. Almost every one of tens of thousands of anchors is background, so the summed loss over easy negatives swamps the gradient from the handful of positives. **Focal loss exists precisely to fix that** — it down-weights confidently-classified examples so the easy negatives contribute little — and it is best understood as a targeted fix for a structural problem rather than a general-purpose improvement to cross-entropy.

2 Non-maximum suppression

```
sort detections by confidence, descending
for each detection, highest confidence first:
    keep it
    discard every lower-confidence detection OF THE SAME
    CLASS whose IoU with it exceeds a threshold
```

THE IoU THRESHOLD IS A REAL DESIGN CHOICE:

- too low -> genuinely overlapping objects are merged, and a crowd becomes one person
- too high -> duplicate boxes survive and precision falls

TWO FAILURE MODES, both common in practice:

CROWDS. Overlapping instances of the same class are exactly what NMS cannot distinguish from duplicate detections of one instance -- the information needed is not in the boxes. Pedestrian and cell detection live in this regime permanently.

ELONGATED OR ROTATED objects. Axis-aligned IoU is a poor similarity measure for a diagonal object, so two boxes around genuinely separate objects can have high IoU and one gets suppressed.

SOFT-NMS decays confidence rather than discarding, which helps crowds measurably. Query-based models (section 1) remove the step entirely, which is the cleaner fix.

NMS runs after the model and before the metric, so **its threshold changes your reported numbers without changing the model at all** — which means a comparison between two systems with different suppression settings is not a comparison of models. **State it when reporting**, and tune it on validation data like any other hyperparameter rather than leaving it at a framework default.

3 Measuring detection

MEAN AVERAGE PRECISION, PRECISELY

A detection counts as correct if its intersection-over-union with an as-yet-unmatched ground-truth box of the same class exceeds a threshold, and each ground truth may be matched by at most one detection — so a second overlapping detection of the same object is a false positive, which is what makes §2's suppression necessary.

Sort all detections by confidence, sweep the confidence threshold downward, trace the resulting precision–recall curve, and take its area. That is average precision for one class; the mean over classes is mAP.

COCO's primary metric averages AP over IoU thresholds from 0.50 to 0.95 in steps of 0.05, so it rewards precise localisation and not merely detection. mAP@0.5 is far more forgiving, and the two numbers are not comparable — typically differing by nearly a factor of two on the same model.

And mAP is an unweighted average over classes, so a rare class counts exactly as much as a ubiquitous one. A model that fails completely on one class out of twenty loses 5 mAP points whether that class appears in every image or in three of them — which is sometimes what you want and frequently not, and either way should be a deliberate choice.

- **Comparing mAP@0.5 against mAP@[0.5:0.95].** The first is roughly twice the second on the same model. Papers report both, readers conflate them, and apparent state-of-the-art jumps sometimes consist of nothing else.
- **Ignoring the object-size breakdown.** Small-object AP is frequently a third of large-object AP (Module 08 §3's resolution argument), so if your application is small objects, the headline number tells you almost nothing. COCO reports the breakdown; read it.
- **Ignoring the operating threshold.** mAP integrates over every confidence threshold, and a deployed system picks exactly one. Report precision and recall at the threshold you will actually use — that pair, not mAP, determines whether the system is usable.
- **Averaging over classes you do not care about,** or being dragged down by a class that is irrelevant to the application — both of which are fixed by reporting per-class AP.
- **And mAP says nothing whatsoever about temporal stability.** A detector that flickers between frames can have excellent mAP and be unusable in video, because flicker is what a user perceives and no per-frame metric measures it.

4 Segmentation

Task	Output	Standard metric
Semantic segmentation	A class label per pixel. Separate instances of the same class are not distinguished.	Mean IoU over classes — and see the caveat below.

Task	Output	Standard metric
Instance segmentation	A mask per detected object, with a class.	Mask AP — §3's mAP with mask IoU replacing box IoU.
Panoptic segmentation	Every pixel assigned both a class and, for countable classes, an instance. The unification of the two above.	Panoptic quality — a recognition term times a segmentation term, which keeps both failure types visible.
Boundary-sensitive evaluation	Any of the above.	Boundary IoU, computed in a band around the contour, because mean IoU hides boundary error almost completely.

Mean IoU is dominated by large regions, since it is computed over all pixels and most pixels are interior. So a model can score very well on mean IoU while getting essentially every boundary wrong — and boundaries are precisely what matters for matting, compositing, and any downstream use that cuts along the mask (CSCE 748 Module 05's requirement). **Report boundary IoU when the boundary is the product.**

CHOOSE THE METRIC FROM THE USE, NOT FROM THE LEADERBOARD

Counting objects? Report count accuracy, not mAP. They diverge sharply: a detector with excellent mAP can systematically miscount, because mAP integrates over thresholds and counting happens at one.

Compositing or matting? Boundary accuracy and alpha error, not region IoU — for the reason just given.

Triggering an action — an alert, a brake, a notification? Precision and recall at your chosen threshold, with the cost of each error type stated, because the two error types almost never cost the same and a single metric cannot express that.

Video? Temporal consistency, which no standard detection metric measures at all. Flicker is the complaint users actually make about video detection systems, and it is invisible to every number in this module. **This is CSCE 633 Module 01 §3's point — the metric is a proxy and the proxy has chosen what you will optimise** — in the specific form it takes in perception.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
Michigan EECS 498 — detection and segmentation lectures (free)	The reference for §1 and §4, with the architecture families laid out clearly.
Lin et al. — Focal Loss for Dense Object Detection (free)	The §1 imbalance problem and its fix, with the analysis that motivates it.

Resource	What it covers for this module
Carion et al. — End-to-End Object Detection with Transformers (DETR) (free)	The matching loss of §1 — read it for the loss, which is the contribution.
COCO detection evaluation documentation (free)	Exactly what the metrics of §3 compute , including the size breakdown that is so often ignored.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Implement mAP from scratch** and verify it against the COCO evaluation tool on the same predictions.
- 2 **Report mAP@0.5 and mAP@[0.5:0.95]** for one model and compute the ratio.
- 3 **Break your results down by object size** and report the three numbers.
- 4 **Sweep the NMS IoU threshold** and plot mAP against it. Report the best value and the spread.
- 5 **Construct a crowded test case** and show NMS merging distinct objects.
- 6 **Try soft-NMS on it** and report the improvement.
- 7 **Pick one confidence threshold** and report precision and recall there. Compare against the mAP.
- 8 **Count objects with your detector** and compare count accuracy against mAP across thresholds.
- 9 **Compute mean IoU and boundary IoU** for a segmentation model and report both.
- 10 **Run a detector on video** and measure flicker — the fraction of frames where a stationary object's detection disappears.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 Why is detection structurally harder than classification?
- 2 Name four architecture families and the trade each makes.
- 3 What was DETR's actual contribution?
- 4 Why does focal loss exist?
- 5 Describe NMS and its two intrinsic failure modes.
- 6 Why does the NMS threshold change your reported metrics?
- 7 Define mAP precisely, including the matching rule.
- 8 Give five ways detection metrics mislead.

- 9 Distinguish the three segmentation tasks and their metrics.
- 10 Why does mean IoU hide boundary error?